

Laboratório 4 – Folha de Dados

Turma: (X) T01 (Segundas) () T02 (Quartas)
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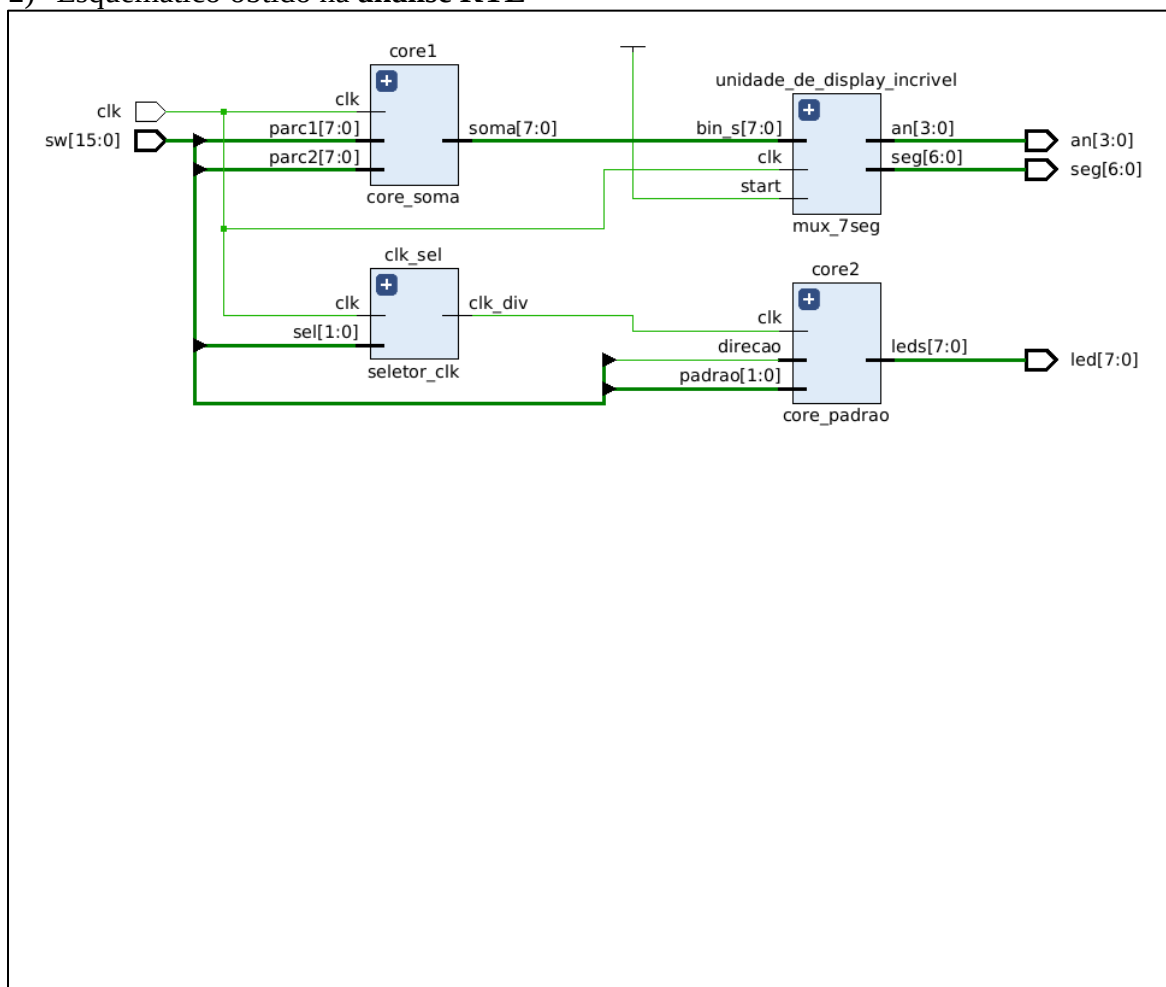
Projeto 1

Circuito RTL usando picoblaze

1) Estimação consumo de recursos lógicos **após a síntese lógica**:

LUTs Total:	FFs Total:	Pinos de IOs Total:	Blocos DSP Total:	Blocos BRAM Total:
308(1.48%)	311(0.75%)	36(33.96%)	--(--%)	2(4.00%)

2) Esquemático obtido na **análise RTL**



3) Códigos Assembly de cada picoblaze

;projeto 1

```
CONSTANT chaves1, 00
CONSTANT chaves2, 01
CONSTANT leds, 02
```

```
function:    INPUT s0, chaves1
             INPUT s1, chaves2
             LOAD s2, s0
             ADD s2, s1
             OUTPUT s2, leds
             JUMP function
```

;projeto_1_sim_2

```
CONSTANT chavesd, 00
CONSTANT chaveps, 01
CONSTANT leds, 04
```

```
function:    INPUT s1, chaveps
             INPUT s0, chavesd

             COMPARE s1, 03
             JUMP Z, pattern3

             COMPARE s1, 02
             JUMP Z, pattern2

             COMPARE s1, 01
             JUMP Z, pattern1

             JUMP pattern0
```

```
pattern3:    LOAD s3, 0D
             COMPARE s4, s3
             JUMP Z, direction
             LOAD s4, s3
             LOAD s5, s4
             JUMP direction
```

```
pattern2:    LOAD s3, 0F
             COMPARE s4, s3
```

```
JUMP Z, direction
LOAD s4, s3
LOAD s5, s4
    JUMP direction
```

```
pattern1:    LOAD s3, 03
             COMPARE s4, s3
             JUMP Z, direction
             LOAD s4, s3
             LOAD s5, s4
             JUMP direction
```

```
pattern0:    LOAD s3, 01
             COMPARE s4, s3
             JUMP Z, direction
             LOAD s4, s3
             LOAD s5, s4
```

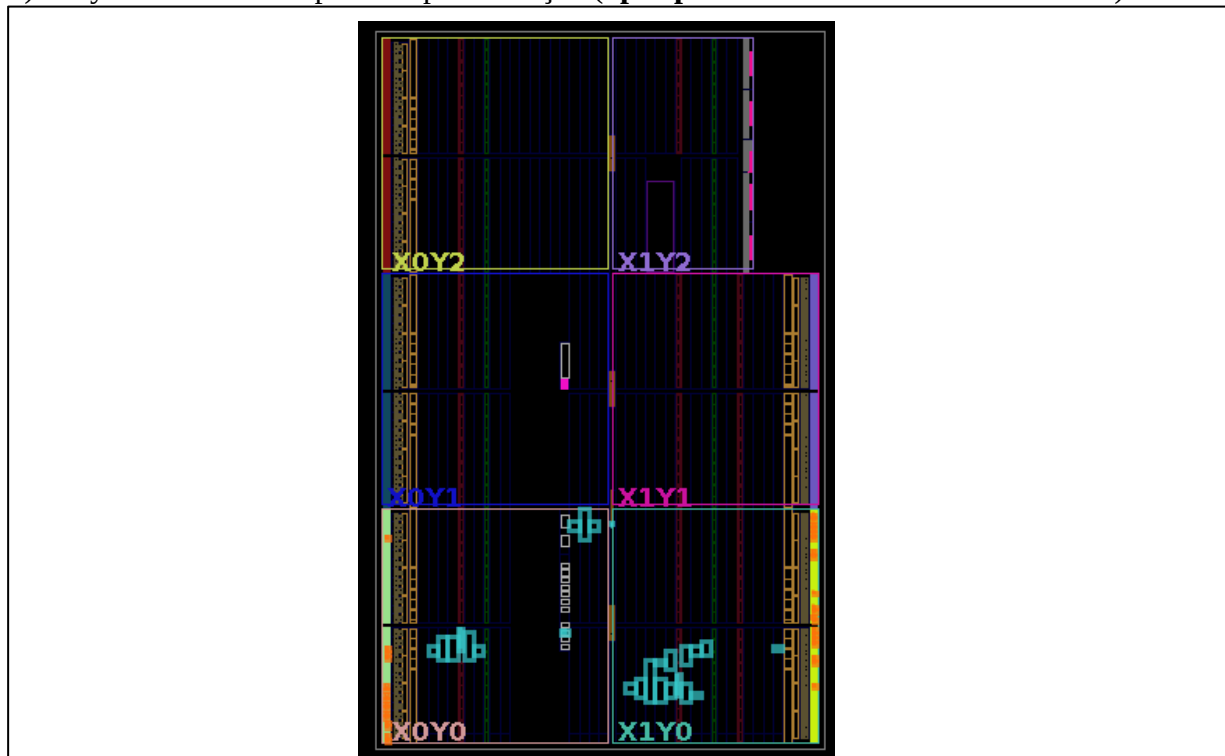
```
direction:   COMPARE s0, 01
             JUMP Z, toleft
             JUMP toright
```

```
toleft:      RL s5
             JUMP done
```

```
toright:     RR s5
             JUMP done
```

```
done:        OUTPUT s5, leds
             JUMP function
```

4) Layout do circuito após a implementação (**após processo *Place and Route* – PAR**):



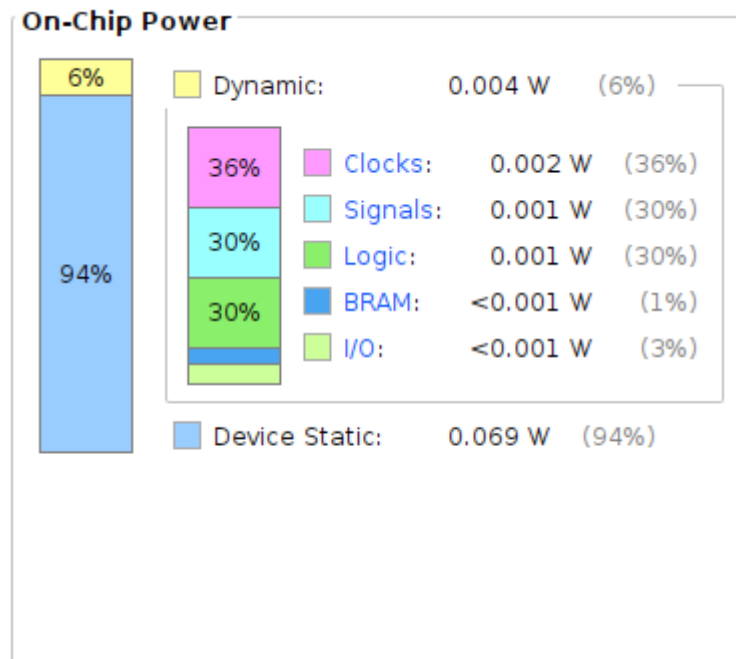
5) Utilização do consumo de recursos lógicos após a **implementação** do circuito:

LUTs Total:	FFs Total:	Pinos de IOs Total:	Blocos DSP Total:	Blocos BRAM Total:
304(1.46%)	310(0.75%)	36(33.96%)	--(--%)	2(4.00%)

6) Estimação do consumo de energia após a **implementação** do circuito:

Potência total: 73(mW)
 Potência estática: 69(mW)
 Potência dinâmica: 4(mW)

Gráfico de consumo de energia:



Timing

Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 1.883 ns	Worst Hold Slack (WHS): 0.101 ns	Worst Pulse Width Slack (WPWS): 3.750 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 606	Total Number of Endpoints: 606	Total Number of Endpoints: 218

All user specified timing constraints are met.

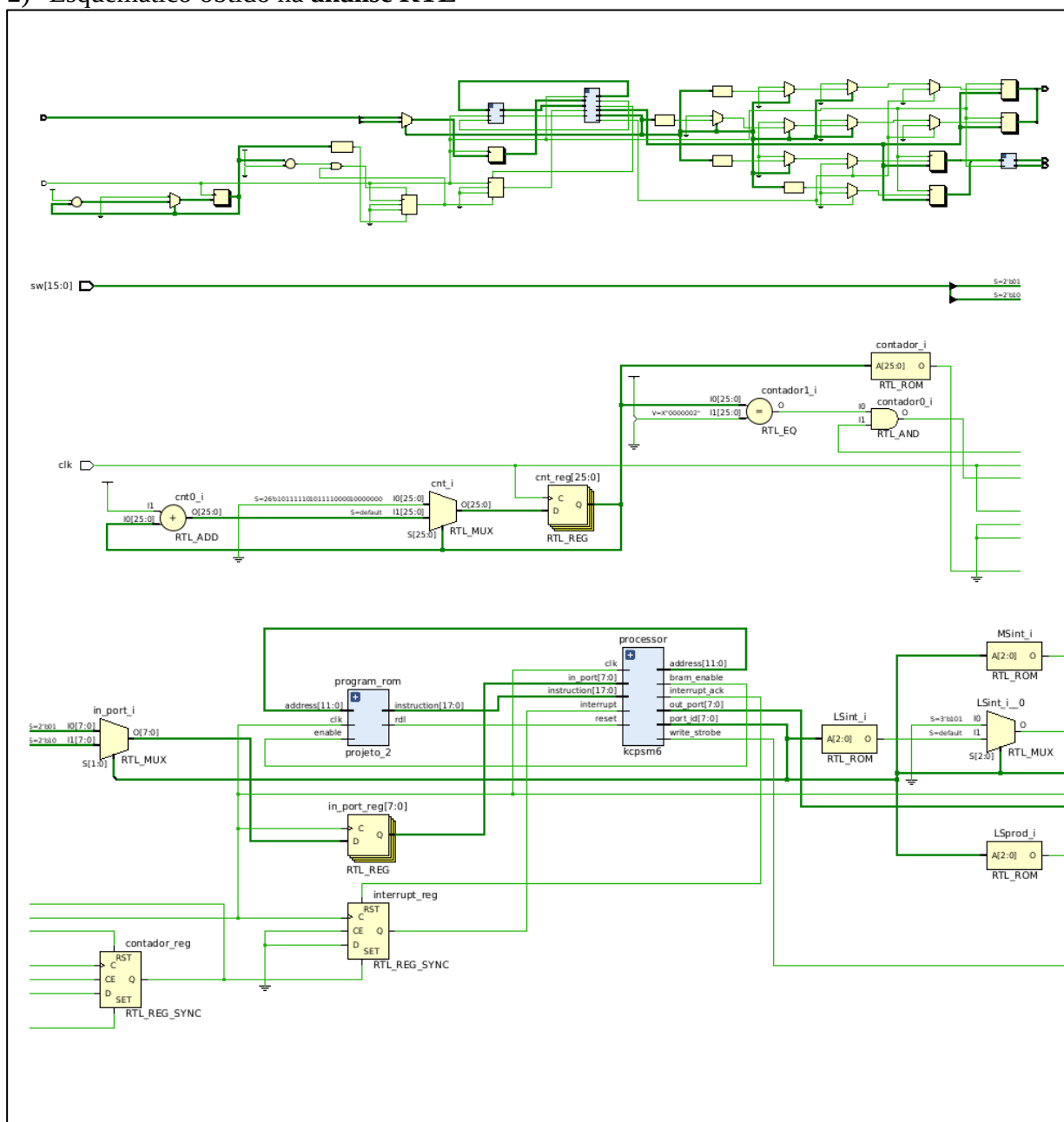
Projeto 2

Circuito RTL usando picoblaze

1) Estimação consumo de recursos lógicos **após a síntese lógica**:

LUTs Total:	FFs Total:	Pinos de IOs Total:	Blocos DSP Total:	Blocos BRAM Total:
167(0.8%)	213(0.51%)	44(41.51%)	--(--%)	1(2.00%)

2) Esquemático obtido na **análise RTL**

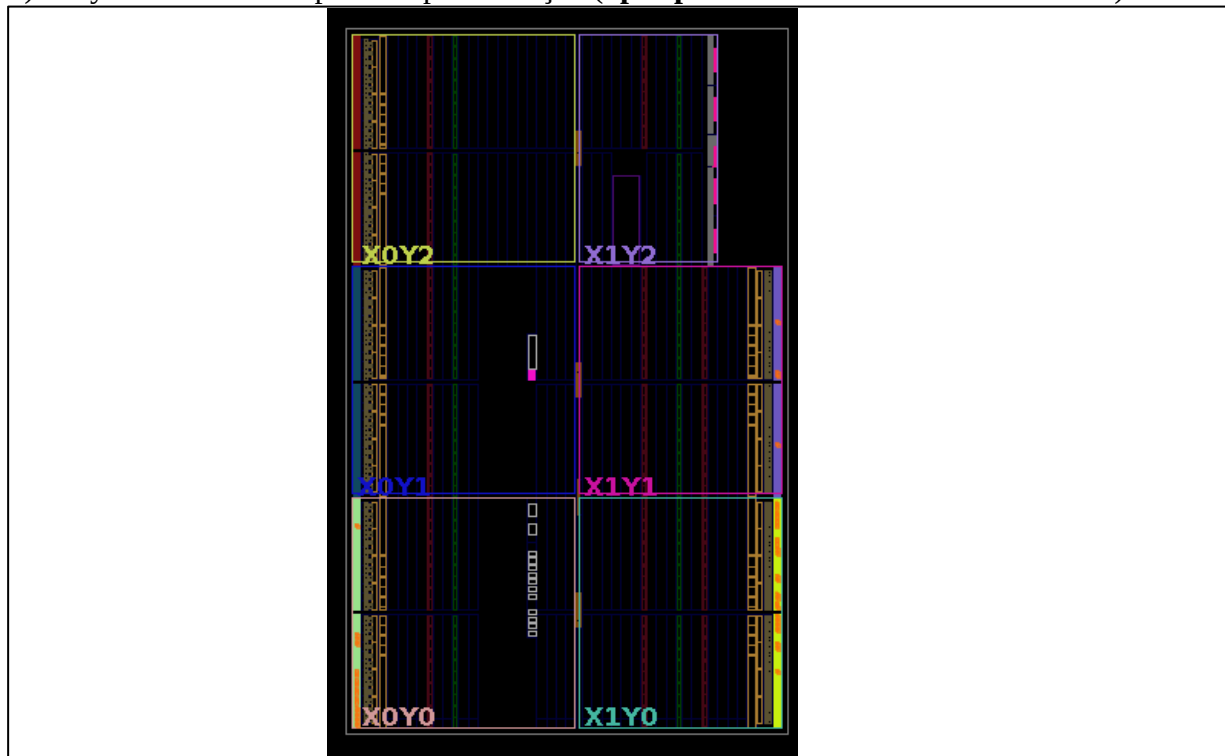


3) Código Assembly do PicoBlaze

```
NAMEREG s1, i           ; indice i mapeado no registrador s1

CONSTANT opA, 01 ; multiplicando de 8 bits na porta 01
CONSTANT opB, 02 ; multiplicador de 8 bits na porta 02
CONSTANT MSprod, 03 ; MS byte do produto na porta 03
CONSTANT LSprod, 04 ; LS byte do produto na porta 04
CONSTANT MSint, 05 ; MS byte do contador de interrupcoes
CONSTANT LSint, 06 ; LS byte do contador de interrupcoes
;
forever:      ENABLE INTERRUPT
               INPUT s3, opA
               INPUT s4, opB
               call mult_soft
               OUTPUT s5, MSprod
               OUTPUT s6, LSprod
               OUTPUT s8, LSint
               OUTPUT s7, MSint
               jump forever
;
mult_soft:    load s5, 00      ; limpa s5, s6 fica em dont care
               load i, 08      ; inicializa indice do laco
;
mult_loop:    sr0 s4           ; desloca a direita s4, carry = LSB
               jump nc, shift_prod
               add s5, s3
;
shift_prod:   sra s5
               sra s6
               sub i, 01
               jump nz, mult_loop
               return
;
ISR:          ADD s8, 01
               ADDCY s7, 00
               returni enable
;
               ADDRESS 3FF
               jump ISR
```

4) Layout do circuito após a implementação (após processo *Place and Route – PAR*):



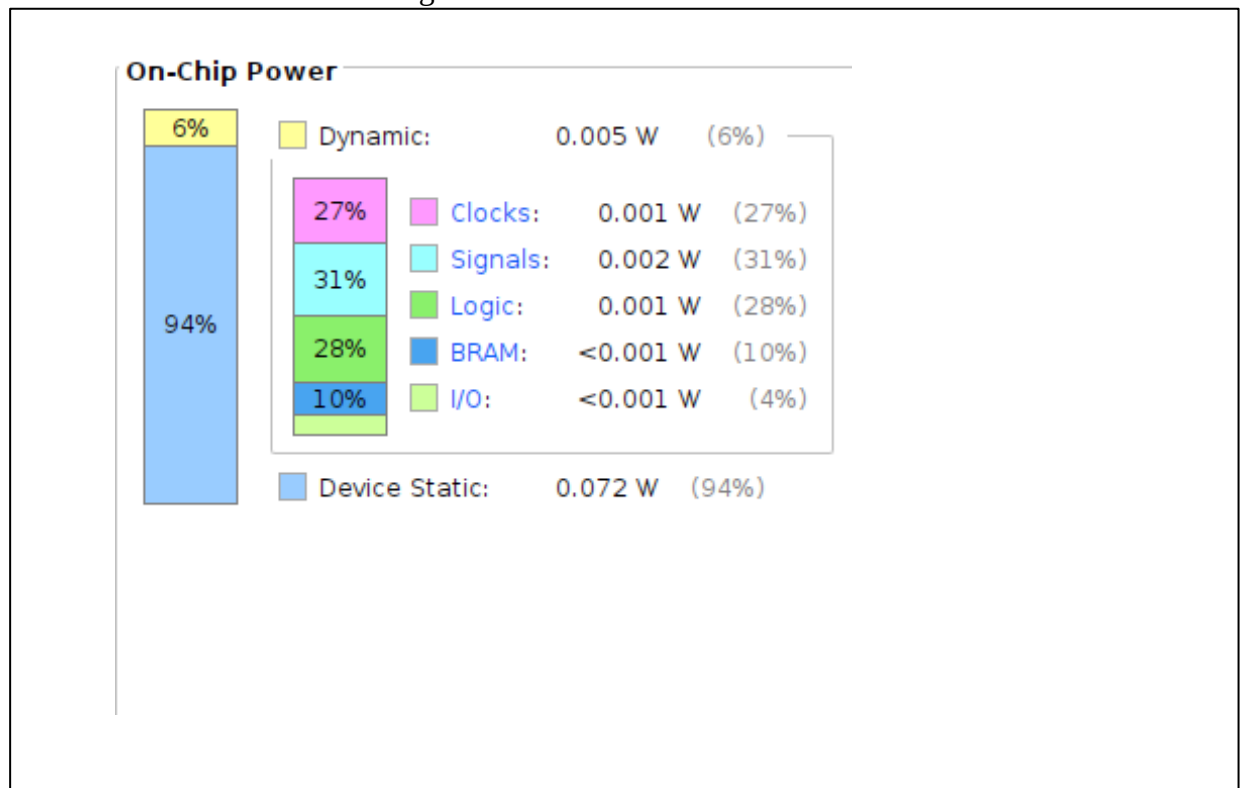
5) Utilização do consumo de recursos lógicos após a **implementação** do circuito:

LUTs Total:	FFs Total:	Pinos de IOs Total:	Blocos DSP Total:	Blocos BRAM Total:
165(0.79%)	213(0.51%)	44(41.51%)	--(--%)	1(2.00%)

6) Estimação do consumo de energia após a **implementação** do circuito:

Potência total: 77(mW)
 Potência estática: 72(mW)
 Potência dinâmica: 5(mW)

Gráfico de consumo de energia:



Timing

Design Timing Summary					
Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	1.979 ns	Worst Hold Slack (WHS):	0.103 ns	Worst Pulse Width Slack (WPWS):	3.750 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	587	Total Number of Endpoints:	587	Total Number of Endpoints:	207
All user specified timing constraints are met.					